

DURANDA ABESINGHE

Full Stack Developer



+94 703796890 "Minsara", Akurugoda, Sulthangoda <https://duranda-abesinghe.github.io/portfolio>
durandaabesinghe@gmail.com <https://github.com/Duranda-Abesinghe>
<https://www.linkedin.com/in/duranda-abesinghe/>

ABOUT ME

Software Engineering undergraduate with hands-on experience in backend development, microservices, and database management. Proficient in C, Java, C#, Python, Kotlin, and HTML. Developed and deployed multiple microservices for academic and personal projects, improving system efficiency and functionality. Experienced in collaborating on team projects to deliver scalable software solutions. Skilled in designing, testing, and optimizing databases and APIs. Continuously learning and applying modern software development practices to solve real-world problems.

PROJECTS

AGRI LINK – DIGITAL PLANTATION & EXPORT MANAGEMENT PLATFORM (2025)

Developed a web and mobile platform serving 500+ workers and 50+ exporters in Sri Lanka. Implemented QR-based inventory tracking across 10 plantations (reducing errors by 60%) and a chat based price system handling 200+ daily transactions. Built real time dashboards for management, improving supply chain decision-making by 40%, using React.js and Firebase with secure role-based authentication and offline support.

AUTONOMOUS WAREHOUSE GOODS MANAGEMENT ROBOT (2024)

Technologies: Arduino, C++, Ultrasonic Sensors, IR Sensors, Bluetooth Responsibilities & Contributions: Designed a system to automatically measure box dimensions (height, width, length) on a conveyor belt using ultrasonic sensors. Displayed delivery location, transmitted via Bluetooth to a robotic vehicle. Developed robotic vehicle for autonomous delivery of boxes to designated locations and returning for the next task. Integrated sensor-based navigation and obstacle avoidance for accurate and safe delivery.

SMART AGRICULTURE – IOT & AI-ASSISTED PRECISION FARMING (2024)

Developed a smart agriculture solution to help smallholder farmers in Sri Lanka improve crop yield and reduce resource wastage.

The system integrates IoT sensors, AI-based decision support, and blockchain concepts to provide real-time recommendations on irrigation, pest control, and supply-chain traceability.

EDUCATION

NIBM SCHOOL OF COMPUTING

Bsc(Hons) Computing
specializing in Software
Engineering
[2023 - 2027]

2024 - Present Higher
National Diploma in
Software Engineering
(Current)

2023 - 2024 Diploma in
Software Engineering
(Completed)

RAHULA COLLEGE-MATARA

[2020-2022]

Advanced Level (Physical
Science Stream)

MAHINDA RAJAPAKSHA COLLEGE-MATARA

[2014-2019]

Ordinary Level

SKILLS

- C
- Java | OOP
- Mobile Application
- HTML | CSS | XML | JavaScript
- MySQL, PHP
- IntelliJ | Visual Studio Code | Canva | Draw.io | Xamp | Wamp
- GitHub
- C#
- Python

CHERIST BISCUIT – OPERATION & QUALITY MANAGEMENT SYSTEM(2024)

Description: Developed a system to streamline production operations and monitor quality control processes for Cherist Biscuit manufacturing.

The system enables efficient tracking of production schedules, inventory management, and quality inspection results, ensuring consistent product standards.

Role & Contributions: Designed and implemented workflows for production and quality monitoring. Created reports and dashboards to track operational efficiency and product quality. Assisted in optimizing inventory and production planning processes.

INVENTORY EXPIRY TRACKER TOOL (2025)

Description: Developed a tool to efficiently manage product inventories by tracking expiry dates using data structures such as Min Heap for sorting and FIFO Queue for stock flow management. The system automatically identifies expired or near-expiry products and sends alerts to prevent losses.

Role & Contributions: Designed and implemented expiry tracking logic, date comparison algorithms, and alert mechanisms. Improved stock handling efficiency, reduced wastage, and optimized overall inventory control.

HOSPITAL MANAGEMENT SYSTEM (2025)

Description: Developed a microservice-based system using Spring Boot (Maven) and MySQL to streamline hospital operations, including doctor management, appointment scheduling, and patient record handling. Implemented authentication modules for doctors and admins to securely manage profiles and appointments.

Role & Contributions: Designed backend microservices for doctor and appointment management. Created RESTful APIs, integrated stored procedures for user registration and login, and optimized database queries to improve system efficiency and scalability.

URBANFOOD E-COMMERCE PLATFORM (2024)

Description: Built an e-commerce platform using Spring Boot and Oracle SQL for backend management, featuring an admin dashboard to manage suppliers, customers, and orders. Integrated MongoDB to handle dynamic user-generated content such as product reviews and feedback.

Role & Contributions: Developed core backend services for admin operations and authentication. Designed database schemas, implemented CRUD operations, and ensured seamless data flow between SQL and NoSQL databases to enhance performance and reliability.

EXTRA CURRICULAR ACTIVITIES

CRICKET

- Captain, Mahinda Rajapaksha College Under-15 & Matara District Team (2018)
- Captain, Mahinda Rajapaksha College Under-17 Team (2019)
- Player, Rahula College Matara First XI Cricket Team (2021 & 2022)

LANGUAGES

- English
- Sinhala (Native)

HOBBIES

- Cricket
- PC-gaming
- Singing

QUALITIES

- Quick Learning & Adaptable
- Team Collaboration
- Leadership Skills
- Organizational Skills
- Time Management
- Critical Thinking
- Accepting Challenges

REFERENCE

- Ms. Chandula Rajapaksha Lecturer / Consultant National Institute of Business Management (NIBM) No. 120/5, Vidya Mawatha, Colombo 07, Sri Lanka +94 70 283 6390 chandula@nibm.lk
- Mrs. Amila De Silva Lecturer / Consultant National Institute of Business Management (NIBM) No. 120/5, Vidya Mawatha, Colombo 07, Sri Lanka +94 71 801 8893 amilau@nibm.lk